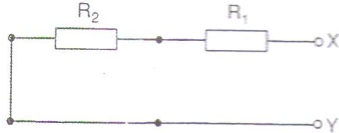


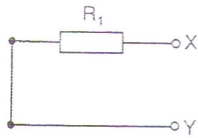
4 (a) $R_1 + R_2 + R_3 = 12 \Omega$

With A open and B closed, resistor R_3 is shorted, the circuit is effectively a series connection of $R_1 + R_2$:



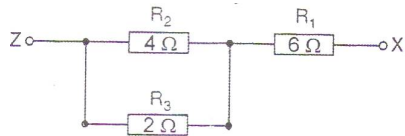
$R_1 + R_2 = 10 \Omega$ and hence $R_3 = 2 \Omega$.

With A closed, both R_1 and R_2 are shorted (regardless of whether B is open or closed), the circuit is effectively just R_3 alone.



- (i) $R_1 = 6 \, \Omega$
- (ii) $R_2 = 4 \, \Omega$
- (iii) $R_3 = 2 \, \Omega$

(b) With A and B closed, the effective resistance between X and Z is effectively:



$$R_{\text{eff}} = \left(\frac{1}{4} + \frac{1}{2} \right)^{-1} + 6 = 7.33 \, \Omega$$